

Statistics Lecture 1

Introduction to statistics

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AALBORG UNIVERSITY
DENMARK

Connectivity

Schedule

Lecture

1. Introduction to statistics
2. Parameter estimation
3. Regression
4. Confidence intervals
5. Hypothesis testing pt. 1
6. Hypothesis testing pt. 2

Workshop: Wrap-up and exam problems

When

March 12

March 24

March 26

March 31

April 7

April 9

TBD

Outline

1. Definition and classification of statistics
2. The law of large numbers
3. The normal (Gaussian) distribution
4. The central limit theorem

Sanity check

What is an event?

Something that can happen. A subset of the possible outcomes of an experiment

What is a probability?

Likelihood or chance of an event happening. Number $[0,1]$. Should sum to 1

What is a random variable?

Mathematical concept that represents an experiment with random outcomes. Outcome to prob.

What is a probability distribution?

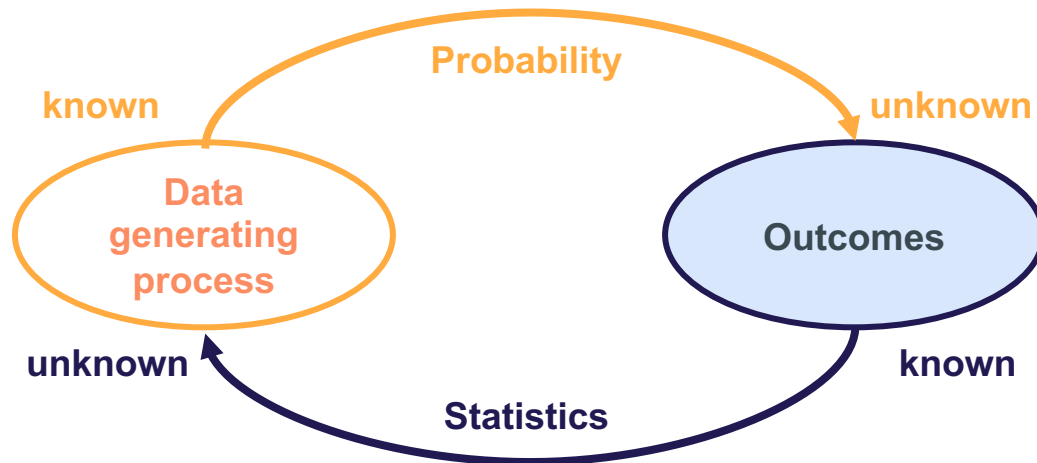
Function that maps possible outcomes to probabilities

1. Definition and classification of statistics

What is statistics?

Probability

Methods to say something about the possible outcomes of a process from a distribution



Statistics

Methods to say something about a process from the outcomes

What is statistics?

Methods to say something about a process from its outcomes

Basis for data-driven methods and machine learning

Statistical inference:

Process of using data samples to infer the distribution that generated them

- Estimation
- Prediction
- Classification
- Clustering

Problem: In everyday statistics, the uncertainty is oftentimes hidden

It's difficult to deal with uncertainty because we are used to live based on certainty

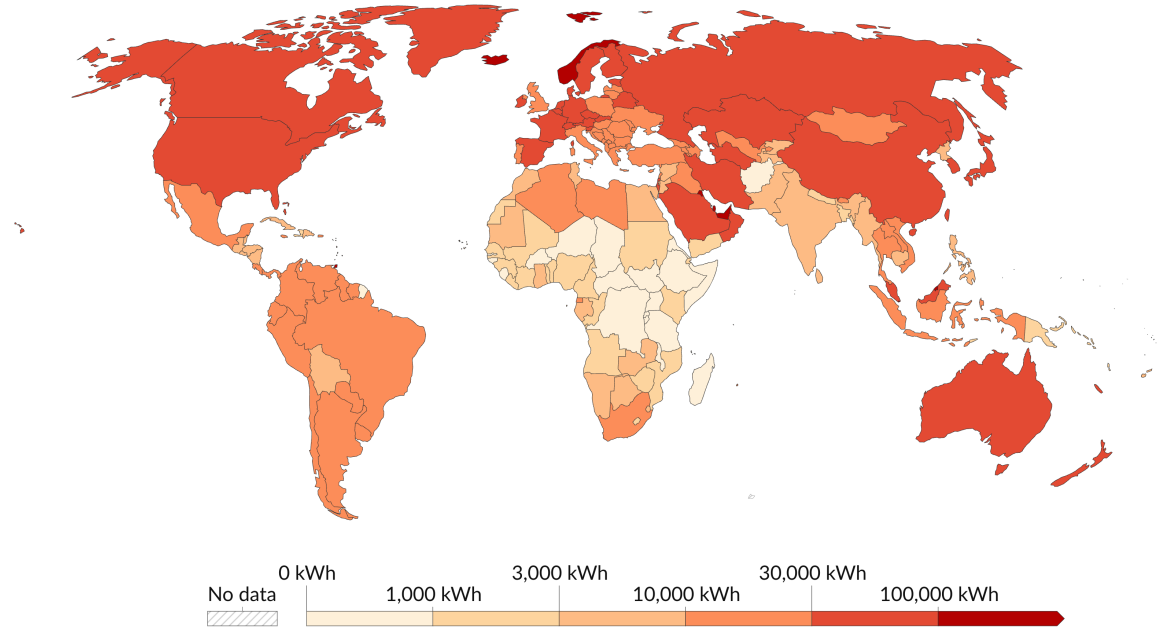
Example

Energy use per person, 2023

Measured in kilowatt-hours¹ per person. Here, energy refers to primary energy² using the substitution method³.

Our World
in Data

Denmark: 33 078 kWh



Data source: U.S. Energy Information Administration (2023); Energy Institute - Statistical Review of World Energy (2024); Population based on various sources (2023)

OurWorldinData.org/energy | CC BY

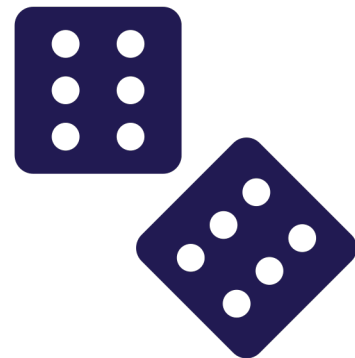
Fair dice example

A fair dice should have the exact same probability of rolling any of the numbers

Can you say if a dice is fair by rolling it once?

Twice?

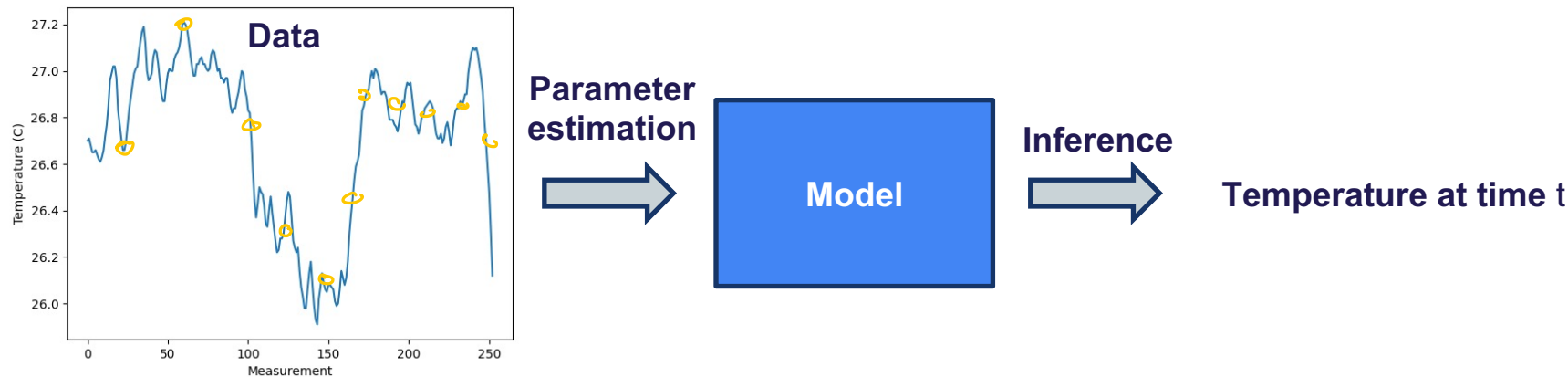
After rolling it 100 times, how confident are you that it is fair?



Why do we need statistics?

In real life, we rarely know the distribution that generates the data points
We might only have some initial assumptions

Statistics allows us to interpret and use the data

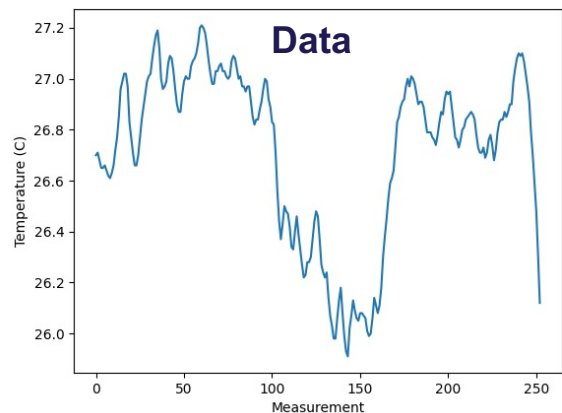


Relation with machine learning

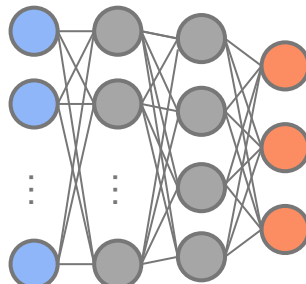
Machine learning is based on statistics

Deep learning exploits a property of neural networks:

They are universal function approximators



Training



Inference



Temperature at time t

Uses of statistics

The scientific method often relies on statistics for empirical confirmation

Descriptive statistics: Describe the data

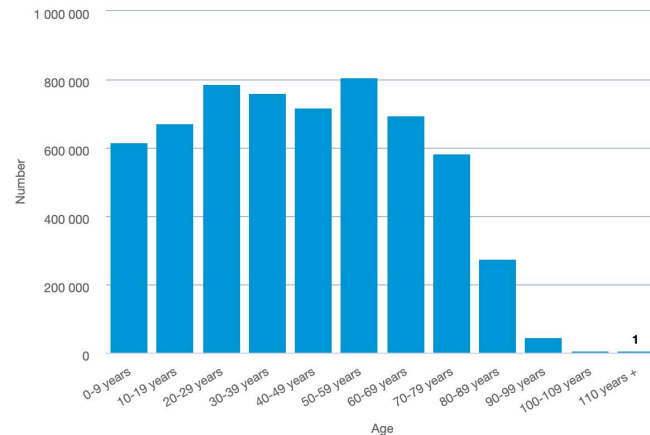
Parameter estimation: Basis to build a model for the data

Hypothesis testing: Confirm or reject an idea

Descriptive statistics:

The population disaggregated by age

Time: Jan 2024:



Source: Statistics Denmark

<https://www.dst.dk/en/Statistik/emner/borgere/befolkning/befolkningstal>

Israel Leyva-Mayorga, Introduction to Probability Theory and Statistics, Spring 2026.



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Descriptive and inferential statistics

Descriptive statistics

Used to present the data in an understandable way

How can we show our results to say something meaningful?

From measurements to graphs, plots, charts, ...

Inferential statistics

Used to extend the results to a larger population

What can we say about the phenomena from the data we have?

From measurements to predictions

Frequentist and Bayesian statistics

Frequentist statistics

Look only at the data, with fixed parameters

Easy to compute

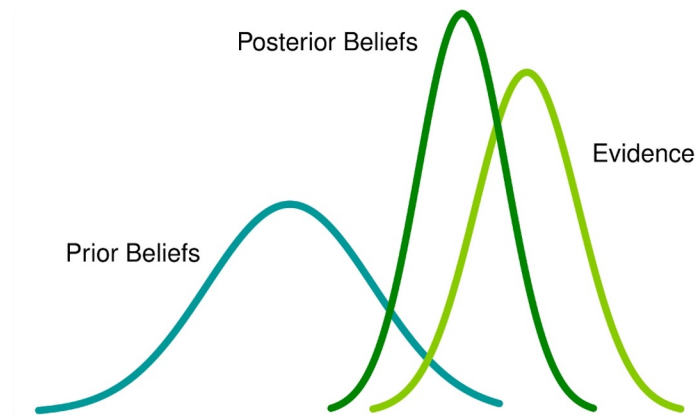
Bayesian statistics

Can include prior knowledge

Can be updated with new information

Parameters are stochastic

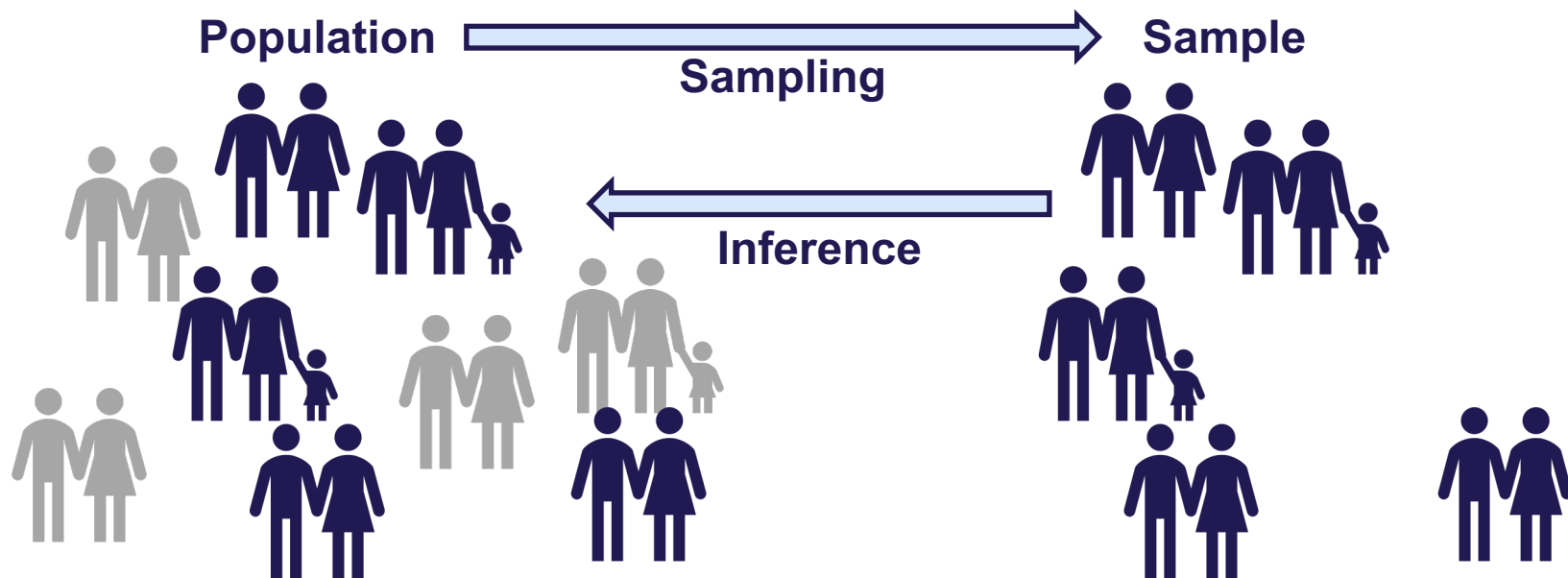
Hard to compute



Sampling

Sampling

If we cannot measure the whole population, we use a smaller sample



Sample space and events

Sample space Ω is the set of possible outcomes of an experiment

Outcomes are individual points $\omega \in \Omega$

Events are subsets of outcomes $A \subseteq \Omega$

Exercise

If we toss a coin twice, where the outcome of each toss is either H or T, then

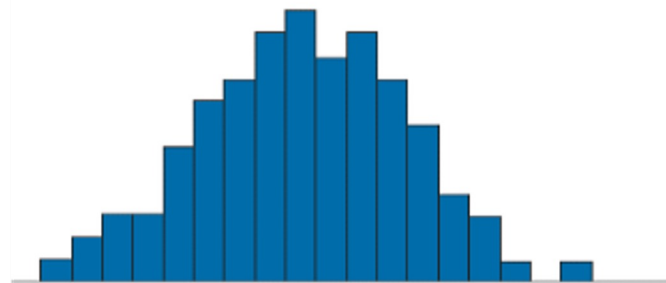
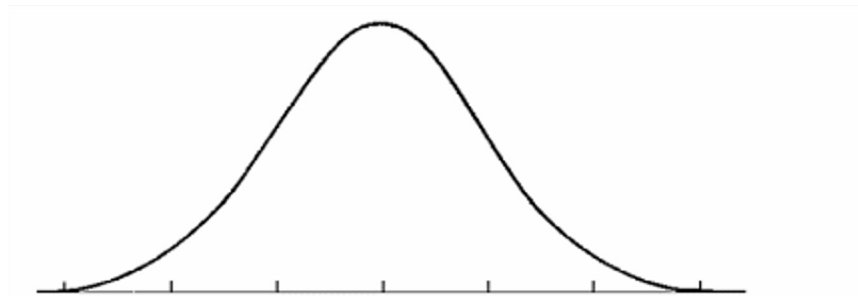
- The possible outcomes are: HH, TT, TH, HT
- The sample space is: $\{HH, TT, TH, HT\} = \Omega$
- The event that the first toss is heads is:
 $A = \{HH, HT\} \subseteq \Omega$

How do we create a sample?

We randomly draw n values from the population

Each value X_i is a random variable with distribution F

We use the sample to estimate some parameter of F (inference)



Implications and assumptions during sampling

If we randomly draw n values from the population

And each value X_i is a random variable with distribution F

Then a sufficiently large sample will ***look like*** the whole population

What does it mean *looking like*?

The parameters of the population are similar to the statistics

The parameters describe the population and the statistics describe the sample

- Mean
- Variance
- Quantiles

Descriptive statistics

Types of data

Quantitative

Objective measurements about some value: age, height, GPA

Qualitative

Categorical data for classification into groups

Name	Age	Gender	Height	GPA	Degree
Andy Smith	21	M	1.83	94.5%	Electronics
Mark Young	23	M	1.77	93.8%	CS
Alice Brown	19	F	1.79	98.2%	Robotics
Jenny Black	22	F	1.64	94.3%	CS
Luke Scott	18	M	1.92	99.1%	Robotics

Summary representation

Absolute frequency of occurrence

How many samples fall in a category?

Relative frequency of occurrence

Frequency normalized by sample size

Must sum to 1 or 100%

Degree	Frequency	Relative frequency
CS	31	$100(31/135) = 23\%$
Electronics	23	$100(23/135) = 17\%$
Robotics	44	$100(44/135) = 33\%$
Architecture	37	$100(37/135) = 27\%$
Total	135	100%

Histograms

A way to show how often we observe different values

1. Divide the range of values into smaller ranges called **bins**
2. Count how many values fall in each bin
3. Make a bar plot with one bar for each bin

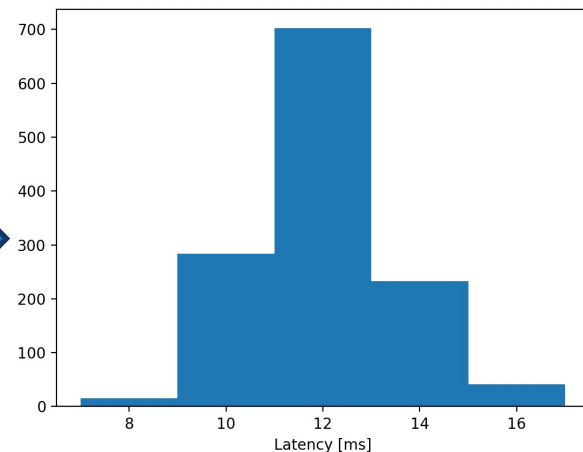
Can be absolute or relative

Main challenge

How to choose the bins?

Bin width

Bins	Frequency
8 – 10	15
10 – 12	284
12 – 14	702
14 – 16	233
16 – 18	41



Histograms

A way to show how often we observe different values

1. Divide the range of values into smaller ranges called **bins**
2. Count how many values fall in each bin
3. Make a bar plot with one bar for each bin

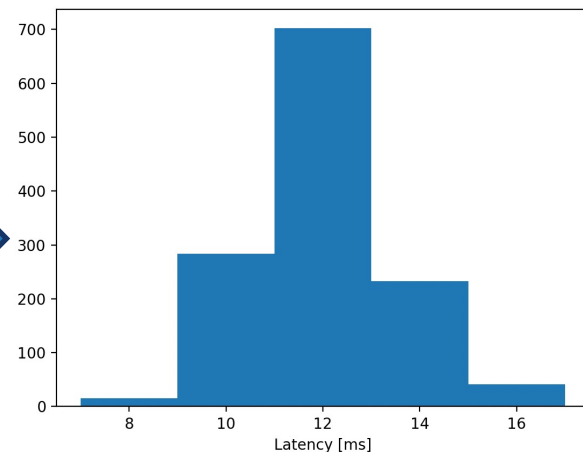
Can be absolute or relative

Main challenge

How to choose the bins?

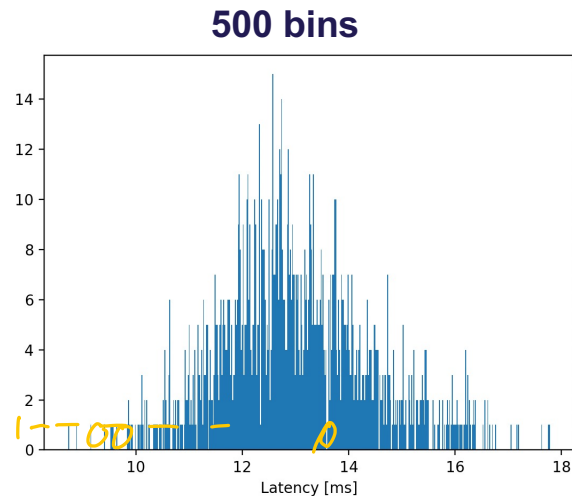
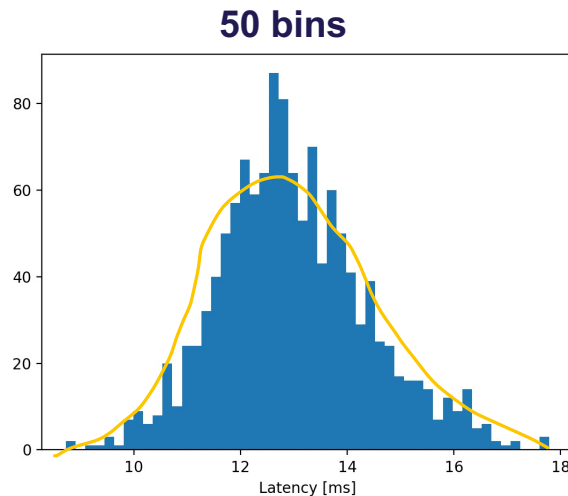
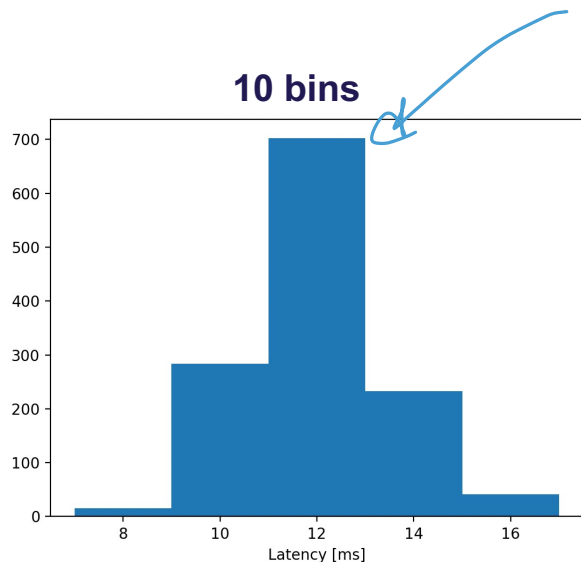
Bin width $w = b^{(i+1)} - b^{(i)}$

Bins	Frequency
$b^{(1)} - b^{(2)}$	$n^{(1)}$
$b^{(2)} - b^{(3)}$	$n^{(2)}$
$b^{(3)} - b^{(4)}$	$n^{(3)}$
$b^{(4)} - b^{(5)}$	$n^{(4)}$
$b^{(5)} - b^{(6)}$	$n^{(5)}$



Example

Execution time (latency) of an image processing algorithm



Types of histograms

Absolute:

Relative: You get an approximation of a pmf

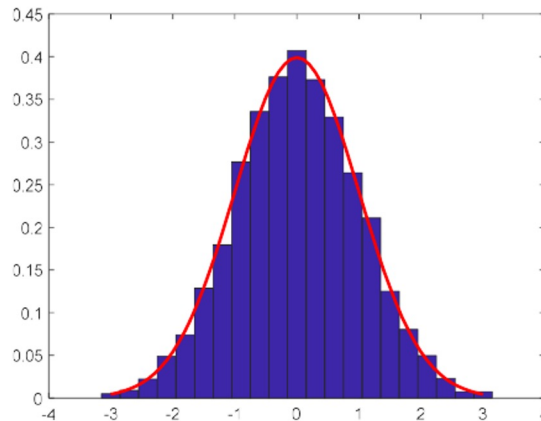
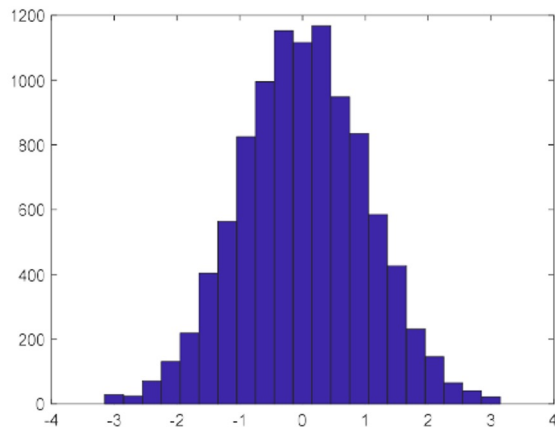
Normalized (by bin size): You get an approximation of a pdf $y^{(i)} = n^{(i)} / (n \times w)$

$$y^{(i)} = n^{(i)}$$

$$y^{(i)} = n^{(i)} / n$$

$$y^{(i)} = n^{(i)} / (n \times w)$$

```
Data = randn(10000,1);
```



$$n = \sum_{i=1}^{\infty} n^{(i)}$$

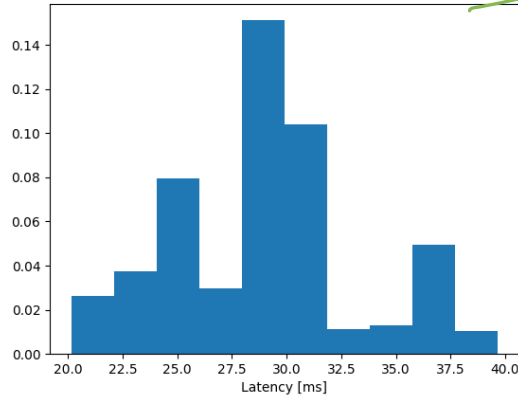
Exercise: Absolute and relative histograms

Which one is which?

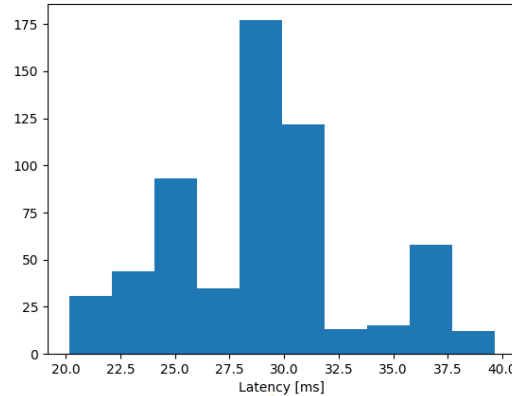
Absolute histogram

Relative histogram

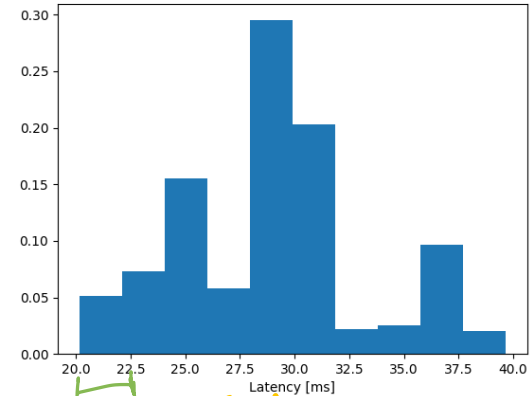
Normalized histogram



(a)



(b)



(c)

Cumulative frequency plot

Back at 9:22

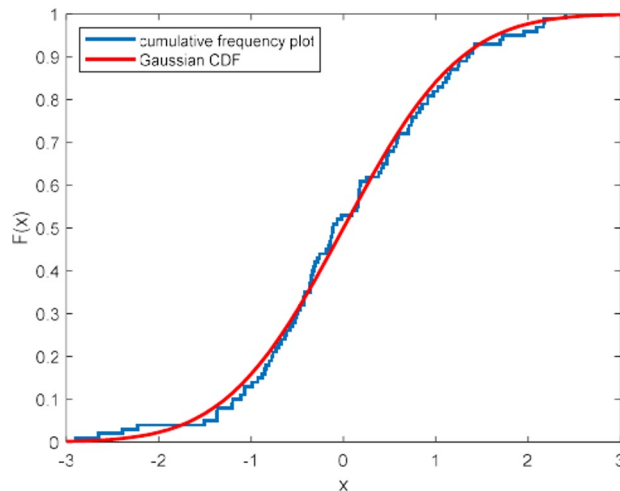
We know that a CDF shows $\Pr(X \leq x)$

Obtained by a cumulative sum of the pmf (relative histogram)

1. Sort the categories or values (ascending order) $b^{(i)} < b^{(i+1)}$
2. The value for a given $x^{(i)}$ is the number of outcomes lesser than $x^{(i)}$

$$F(x^{(i)}) = \frac{1}{n} \sum_{j=0}^i n^{(j)}$$

Plot $\{(x^{(i)}, F(x^{(i)}))\}$



Sample mean and variance in descriptive statistics

If $x_1, x_2, \dots, x_n$ are the samples, the **sample mean** is

$$\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$$

And the **sample variance** is

$$s_n^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)^2$$

Sample mean and variance in inferential statistics

If $X_1, X_2, \dots, X_n$ are random variables, **the sample mean is the random variable**

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

And the **sample variance is another random variable**

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Important

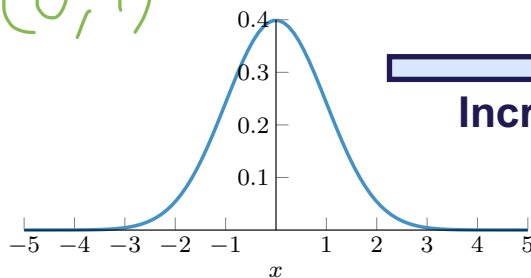
The square root of the variance is the **standard deviation**

Look at the denominator in the formula for the sample variance

Sample mean and variance visualized

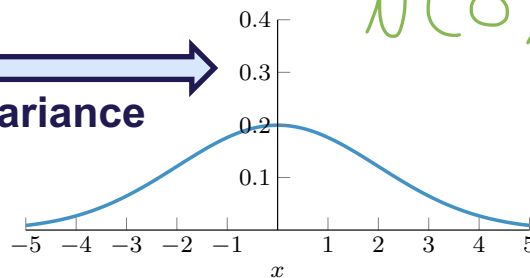
$$\sigma^2 > 1$$

$$N(0, 1)$$



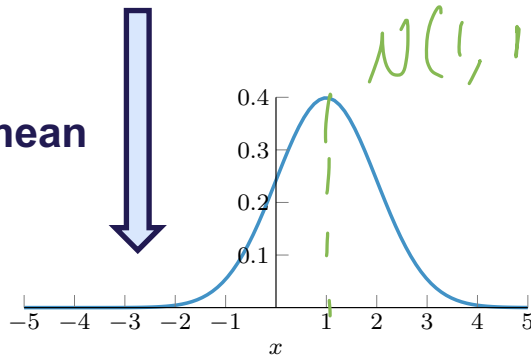
→
Increase variance

$$N(0, \sigma^2)$$



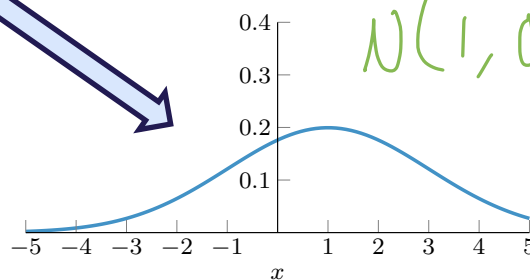
↓
Increase mean

$$N(1, 1)$$



↘

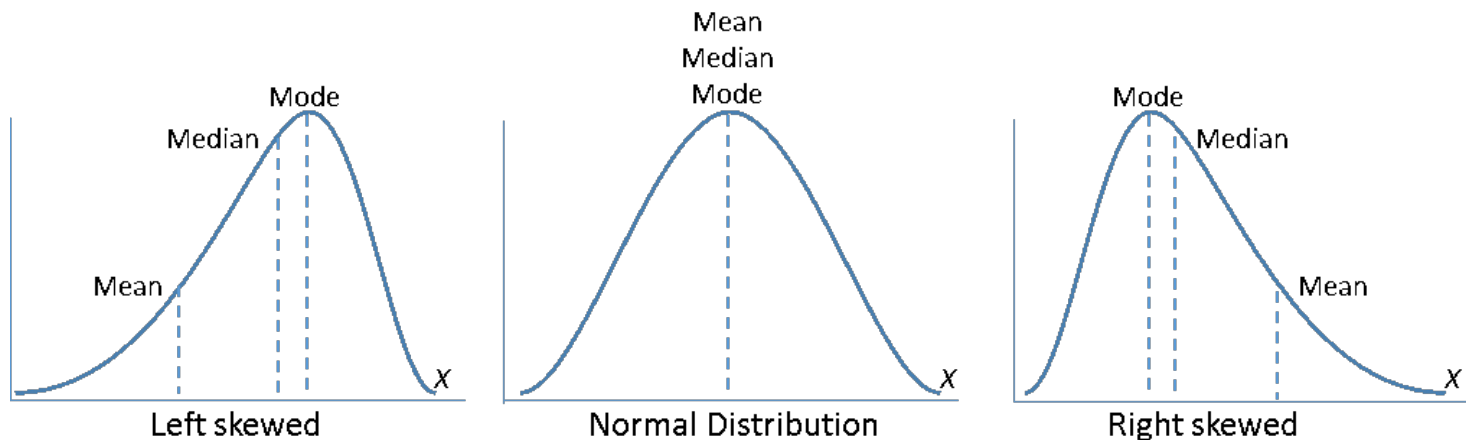
$$N(1, \sigma^2)$$



The mean is not always equal to the median

Median: 50% of the outcomes are below and 50% are above

Mode: Category with highest frequency



The law of large numbers

The linearity of the expectation and sample mean

Expectation

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$$

Sample mean

sample mean

$$\text{If } \underline{Z_i} = X_i + Y_i, \text{ then } \underline{\bar{Z}_n} = \underline{\bar{X}_n} + \underline{\bar{Y}_n}$$

$$\text{If } \underline{Z_i} = \underline{aX_i + b}, \text{ then } \underline{\bar{Z}_n} = \underline{a\bar{X}_n} + b$$

THE LAW OF LARGE NUMBERS

If $\underline{X_1}, \underline{X_2}, \dots, \underline{X_n}$ are i.i.d. random variables and the sample mean is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n \underline{X_i}$$

Then, $\underline{\bar{X}_n}$ **converges in probability** to $\underline{\mu} = \underline{\mathbb{E}(X_i)}$ as $\underline{n \rightarrow \infty}$

$$\underline{\bar{X}_n} \xrightarrow{P} \underline{\mu} = \mathbb{E}(X_i)$$

So, $\bar{X}_n$ is close to μ with high probability if the sample size is large

The normal (Gaussian) distribution

The Gaussian distribution

Also called **normal distribution**

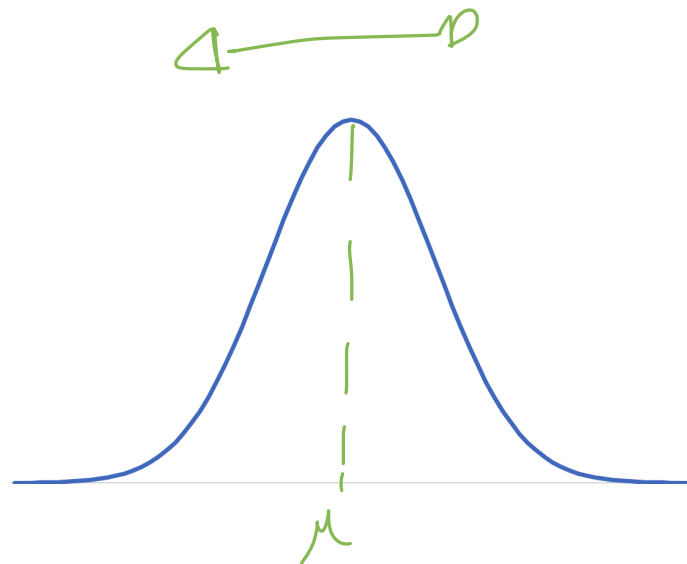
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean μ , located at the center of the distribution

Variance σ^2 tells us how wide it is

Compact representation is $N(\mu, \sigma^2)$

■ If X is normal RV we say $X \sim N(\mu, \sigma^2)$



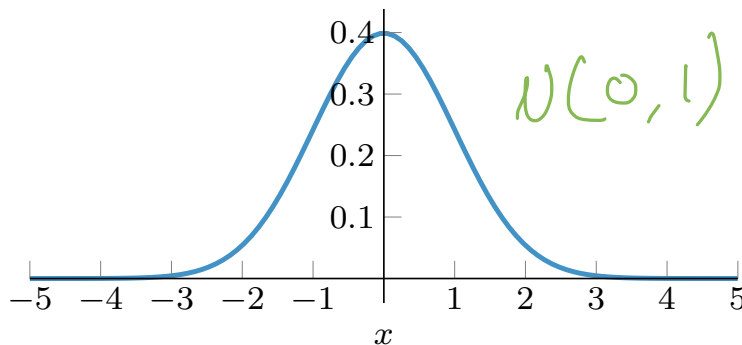
Question:
Is it continuous or discrete?

Standard normal distribution $N(0,1)$

The case of a normal distribution with $\mu = 0$ and $\sigma^2 = 1$
Denoted by small greek letter phi

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Probability distribution function (PDF)



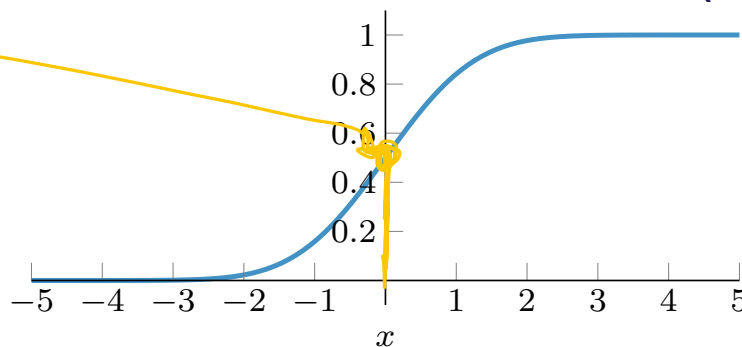
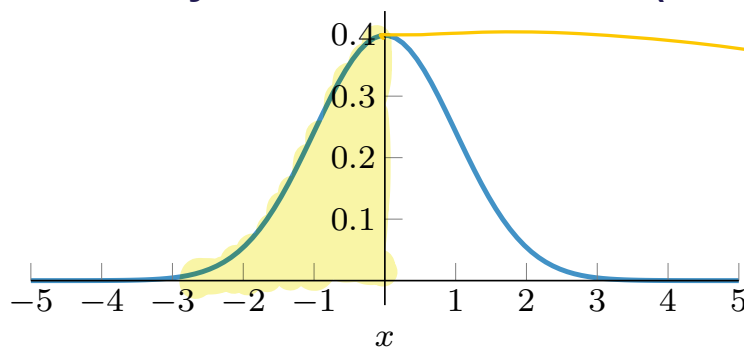
Cumulative Distribution Function (CDF)

Denoted by $\Phi(x)$ (Phi of x)

$$\boxed{\Phi(x)} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right) \right]$$

Probability distribution function (PDF)

Cumulative distribution function (CDF)



Every normal distribution is a scaled standard normal

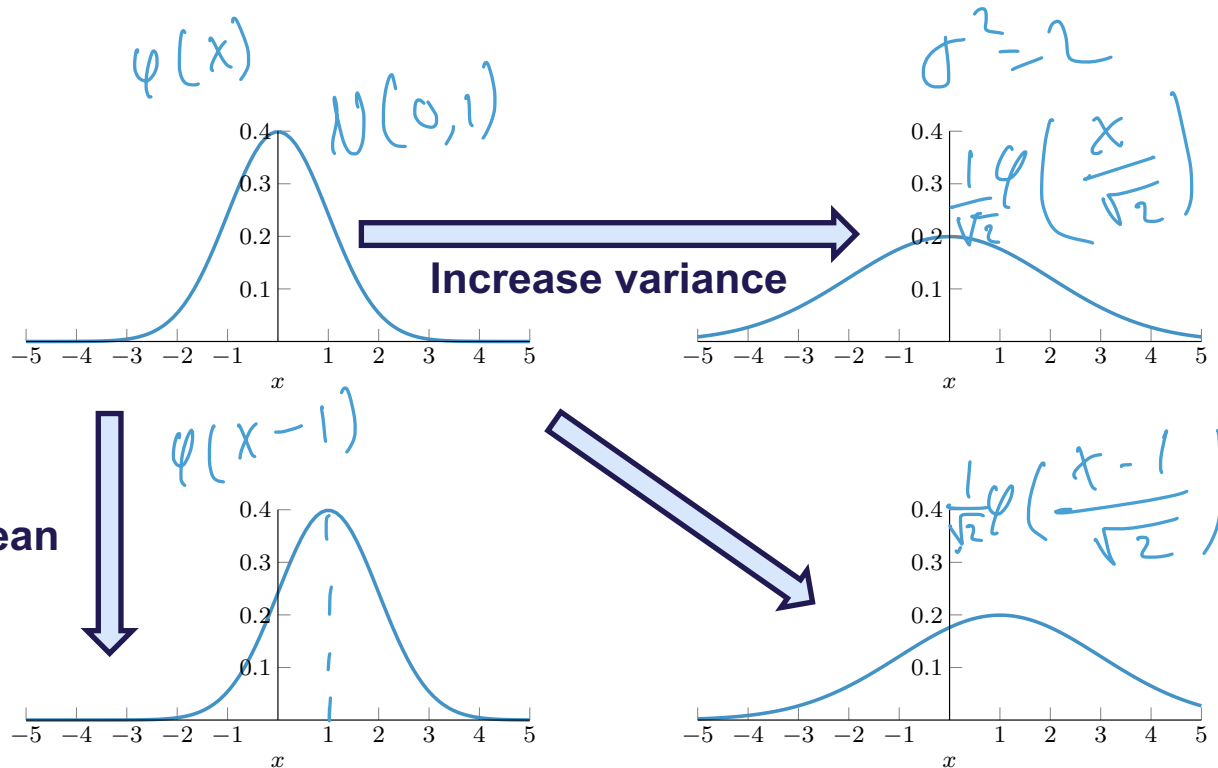
PDF

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma} \varphi\left(\frac{x - \mu}{\sigma}\right)$$

CDF

$$F(x; \mu, \sigma^2) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Increase mean

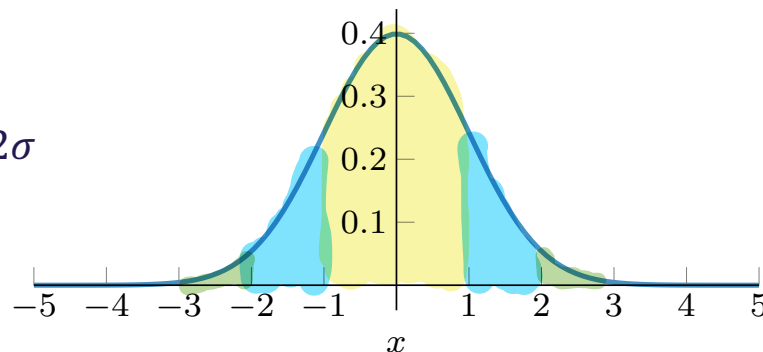


Using the standard normal

$N(0,1)$

Probabilities of the outcomes using σ

- 68% of outcomes are between $\mu - \sigma$ and $\mu + \sigma$
- 95% of outcomes are between $\mu - 2\sigma$ and $\mu + 2\sigma$
- 99.7% of outcomes between $\mu - 3\sigma$ and $\mu + 3\sigma$



Probability that the outcome of a normal RV $X \sim N(\mu, \sigma^2)$ is between a and b

$$\begin{aligned} P(a < X < b) &= F(b; \mu, \sigma^2) - F(a; \mu, \sigma^2) \\ &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{b - \mu}{\sigma\sqrt{2}}\right) \right] - \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{a - \mu}{\sigma\sqrt{2}}\right) \right] \end{aligned}$$

Linear operations with the normal distribution

If $X_i \sim N(\mu_i, \sigma_i^2)$ and $i = 1, 2, \dots, n$ are independent, then

$$\sum_{i=1}^n X_i \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

If $X \sim N(\mu, \sigma^2)$, then $aX + b$ has mean $a\mu + b$ and standard deviation $|a|\sigma$

variance $a^2 \sigma^2$

Any linear combination of normal RVs X_1 and X_2

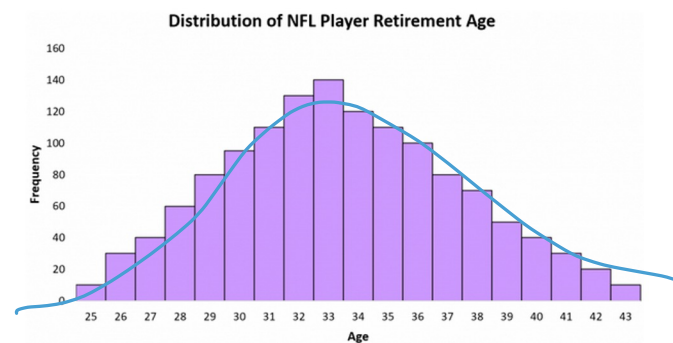
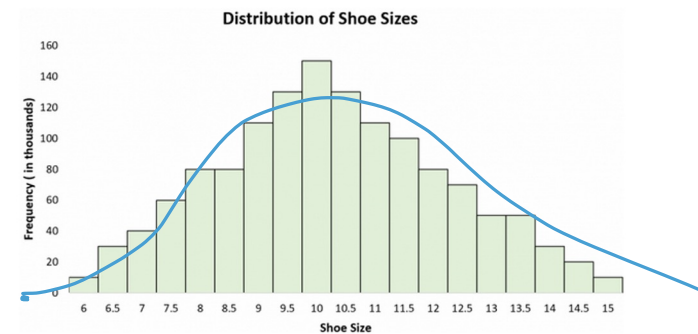
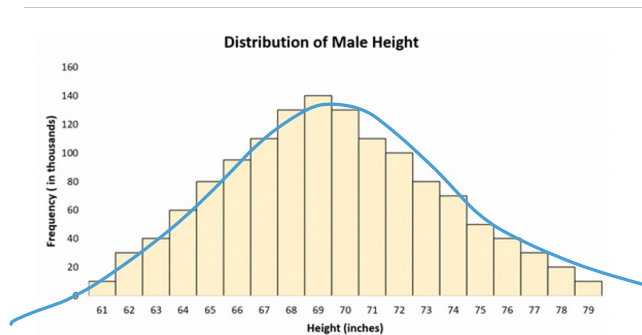
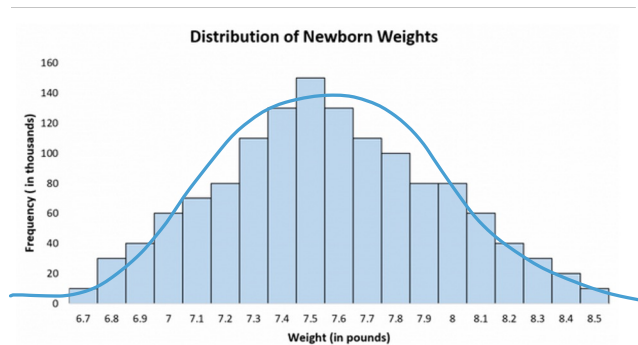
$$W = aX_1 + bX_2$$

has a normal distribution $W \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$

Check normal.py

The central limit theorem

Normal distributions are everywhere



THE CENTRAL LIMIT THEOREM

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with mean μ and variance σ^2 .
Then, if n is large:

$$\sum_{i=1}^n X_i = X_1 + X_2 + \dots + X_n \approx N(n\mu, n\sigma^2)$$

In words

Regardless of the distribution of $X_1, X_2, \dots, X_n$

The sum of n i.i.d. random variables is approximately normally distributed
with mean $\mu_n = n\mu$ and variance $\sigma_n^2 = n\sigma^2$.

Example: Board game

A board game requires the player to roll multiple die
The outcome is the sum of values of the die

How does the distribution look like?

One dice: uniform between 1 and 6

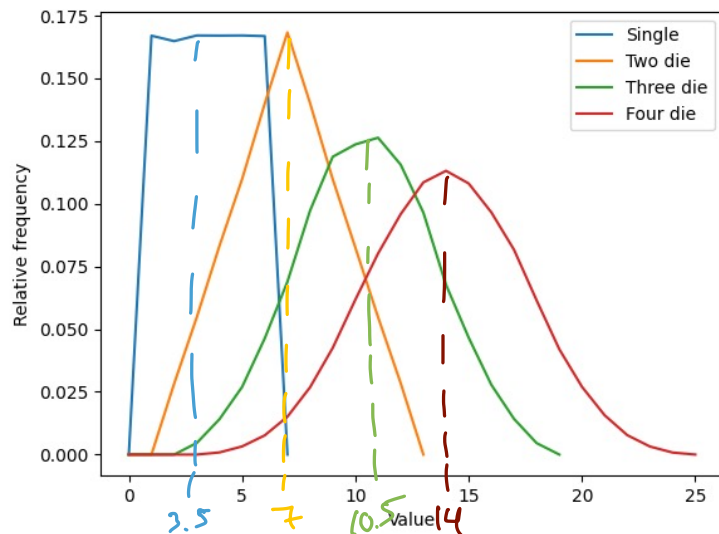
$$X_i \sim U(1,6)$$
$$\mathbb{E}(X_i) = \sum_{k=1}^6 p_k k = \sum_{k=1}^6 \frac{1}{6} k = 3.5$$

Two, three, four...?

The more die the more it looks normal

Mean $\mu_n = 3.5n$ Variance $\sigma_n^2 = \frac{n(6^2-1)}{12}$

$$\frac{(b-a+1)^2-1}{12} = \frac{6^2-1}{12}$$



Exercise: Coin toss

A coin toss X_i is a Bernoulli RV with probability $p = 0.5$

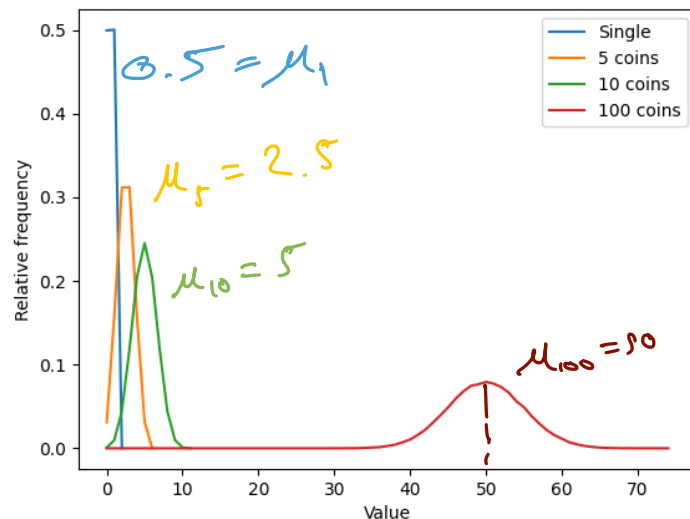
Outcomes are heads H and tails T

Now we toss the coin n times

The number of heads is a binomial RV H_n

Find the mean ~~of~~ μ_n

1. Using the formulas for Binomial distribution
2. Using the Central Limit Theorem
3. Does it match the graph to the right?



Concluding remarks

Lessons learnt

Statistics are everywhere, and it's important to learn them

The basic ways of **describing and visualizing statistical data**:

- Distributions: PDF and pmf
- Cumulative plots: CDF and empirical CDF (ogive)
- Others: box plots, running averages

Sampling is needed and the sample should be representative of the population

Law of large numbers

When the sample is large, we can accurately approximate the mean

The Central Limit Theorem

The sum of a large number of RVs is approximately normal

Useful links and packages

Sources for data:

Our world in data: <https://ourworldindata.org/>

Matlab

Commands `normpdf` and `normcdf` to easily create a normal PDF and CDF

Python

Draw random samples from normal distribution

Numpy: `np.random.normal(loc=mu, scale=std, size=(m,n,k))`

Scipy: `scp.stats.norm.()` creates an RV object with the specified parameters

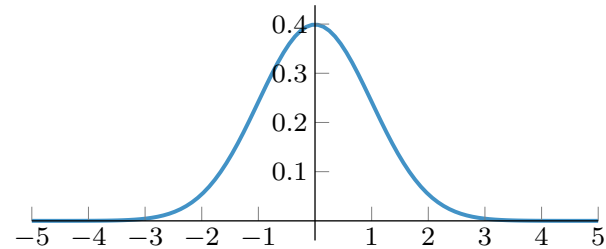
Appendix: Standard normal distribution and Q-function

Cumulative Distribution Function (CDF) is denoted by $\Phi(x)$ (Phi of x)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

Q-function

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-\frac{t^2}{2}} dt$$



Relation to Q-function

$$\Phi(x) = 1 - Q(x)$$

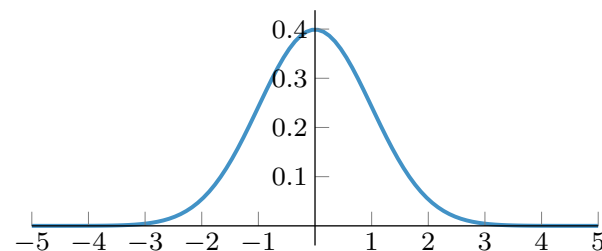
Appendix: Standard normal distribution and error function

Cumulative Distribution Function (CDF) is denoted by $\Phi(x)$ (Phi of x)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

Error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$



Relation to error function

$$\Phi(x) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right) \right]$$